POM: Evaluating model's capability to reproduce multiple patterns

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Objectives

- An ABM opinion dynamics model was developed to reproduce the opinion change during the Mexican 2024 presidential election. It was found to be capable of reproducing the main pattern considered: empirical preference share registered in a selected opinion poll.
- For the current experiments, we follow Pattern Oriented Modeling (POM) [3] and evaluate the model's capability to reproduce multiple patterns observed in an election, such as vote share by gender and age group.

Structural realism

- An ABM should be as simple as possible while still capturing and reproducing the key characteristics and behaviors of the real system, based on the objectives of the simulation.
- This is called *structural realism*: including only the essential components of a real system that are needed to produce the main properties and behaviors.

Pattern Oriented Modeling (POM)

- A model can be capable of reproducing the main pattern observed in a phenomena while not having structural realism.
- POM helps to identify the main components necessary in the model, by using *multiple* observed patterns in the real system. A model capable of reproducing multiple patterns is more likely to have achieved structural realism.

Chosen opinion polls

- The proposed model is based on the study carried over by the group GEA-ISA for the election analyzed [2]. It reports the voting intention of 1070 mexican citizens interviewed at a nationwide scale, registering multiple demographic variables.
- The data was adjusted to consider only the two main candidates. The election results were also considered, and the resulting preference share is as shown in the following table.

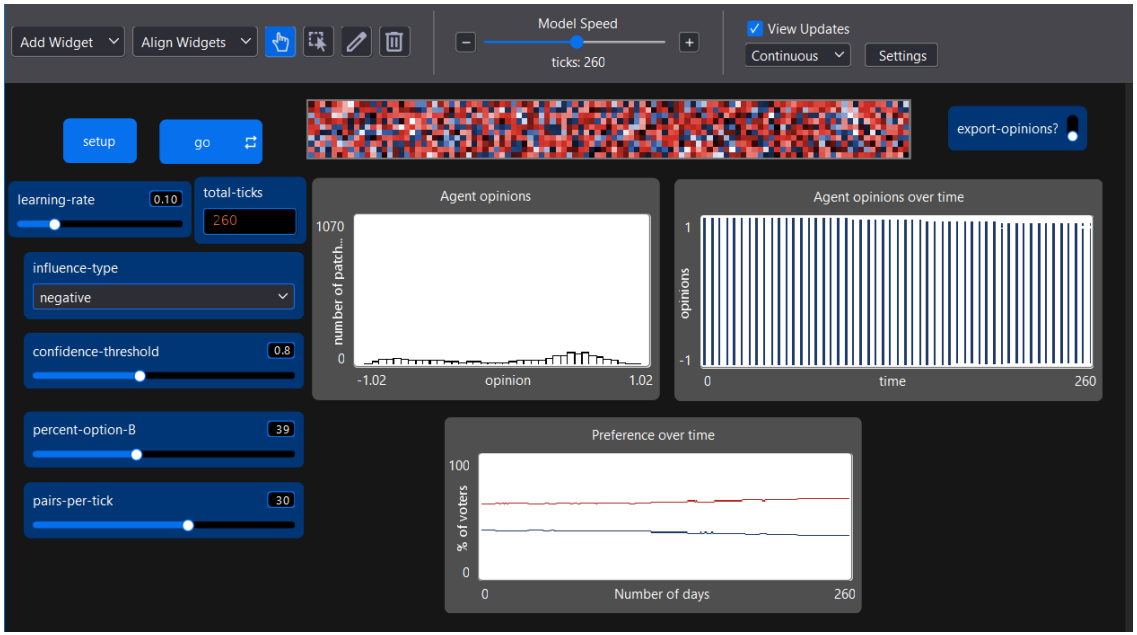
Date	Time (days)	Preference for A (%)	Preference for B (%)
2023/09/16	0	61	39
2023/11/25	70	64	36
2024/03/01	167	61	39
(Election day) 2024/06/02	260	68	32

Basic model

- A basic model of opinion dynamics was developed considering the poll's characteristics and the restrictions of a political election. The model considers three different types of influence, based on a classification given by [1, 4].
- The model consist of 1070 individual, with every time step representing a day. As such, a run consist of 260 time steps or ticks, considering the days since the first poll until the election day. The initial preference share is taken from the first available poll.
- The model was capable of reproducing the main pattern of preference share at the three dates with empirical data.

Limitations of the basic model

- The developed basic model doesn't consider any characteristic for the agents besides their initial preference. As such, the only pattern capable to be validated is the preference share.
- To validate multiple patterns, the model needs to be modified so the agents have multiple characteristics beyond their opinion. This is done considering age and gender.



Obtaining preference share by gender

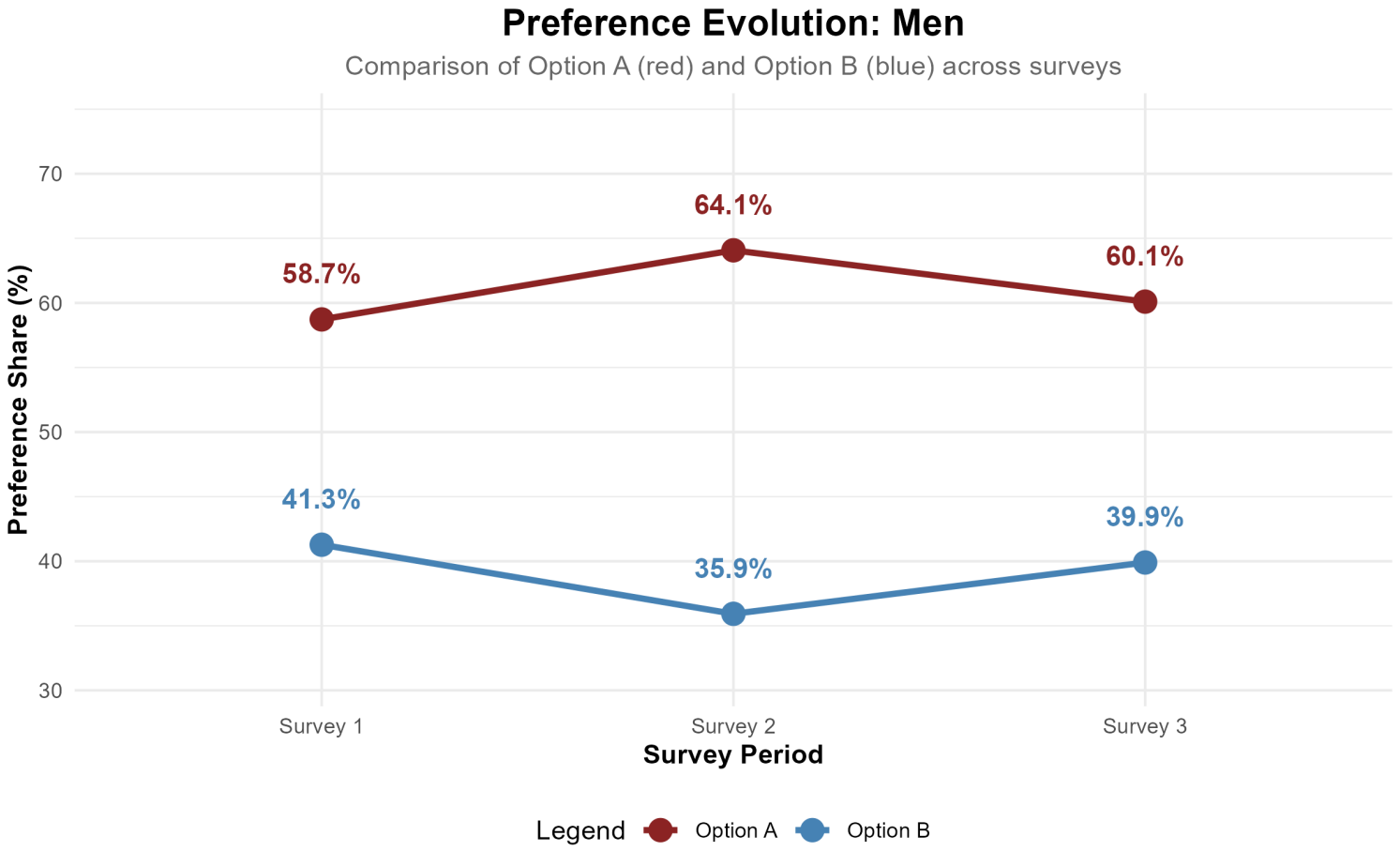
- Survey data is weighted to represent the proportions reported in the national electoral list, and this was taken into account while reducing the data to the two main options.
- The electoral registry shows an electorate composed of 48% men and 52% women.
- The preference share by gender was calculated for each one of the available polls, first dividing the population by gender and then calculating each subgroup preferences.

Evolution of preference share by gender

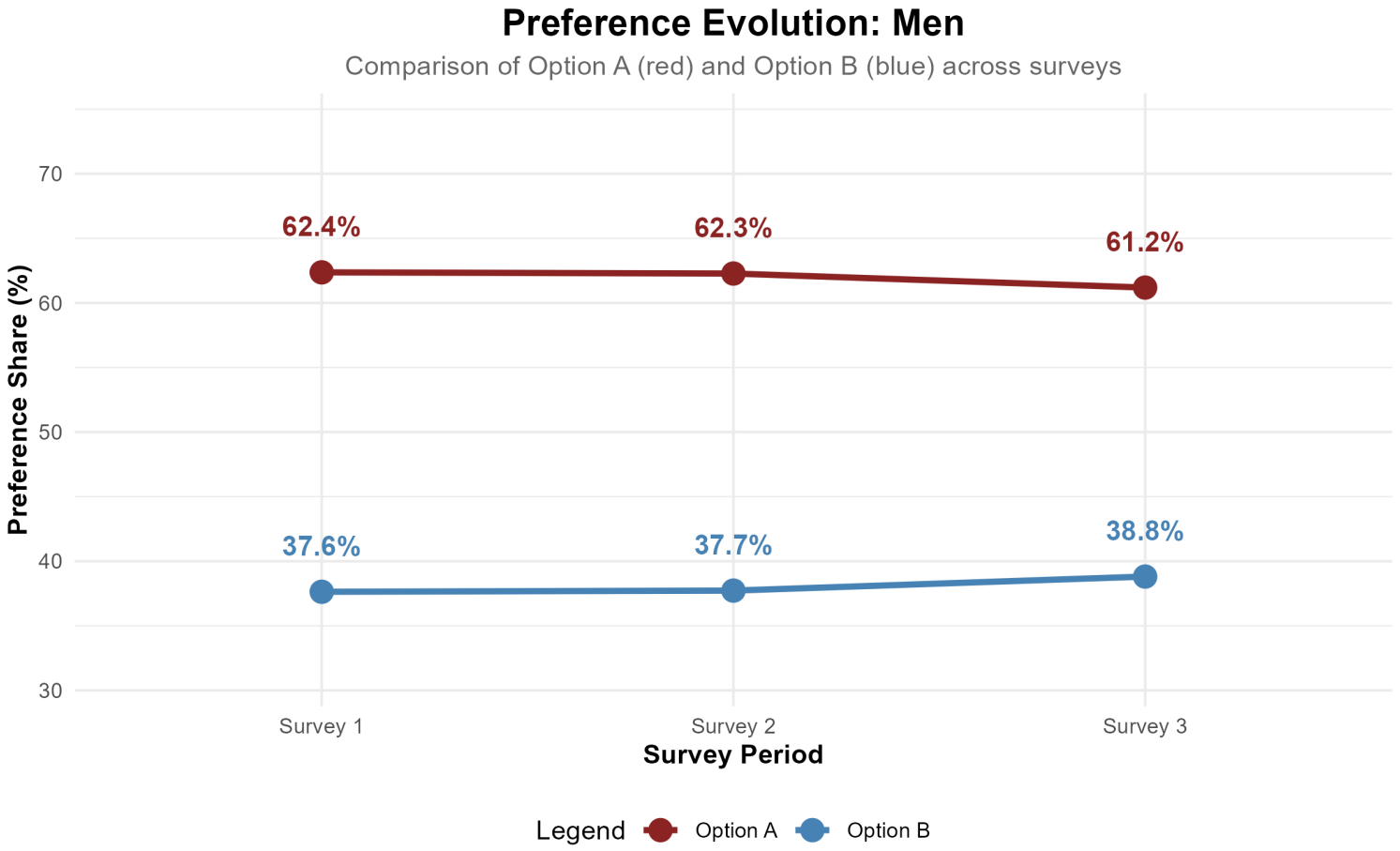
The evolution of the preference share for both genders is shown in the following table. We see almost no change for women, while men show an overall increase of support for B, which we seek to see replicated in the model.

Gender	Option	Survey 1 (%)	Survey 2 (%)	Survey 3 (%)
Men	Option A	58.7	64.1	60.1
	Option B	41.3	35.9	39.9
Women	Option A	62.4	62.3	61.2
	Option B	37.6	37.7	38.8

Evolution of preference share for men



Evolution of preference share for women



Age groups in electorate

The population is divided in four age groups, starting at 17 years old. The age share for the electorate on a nationwide level is shown in the following table, which is replicated in the polls by the use of a weighting factor. This proportion is maintained for the three surveys.

Age group	**% of electorate**
17-24 years old	16
25-39 years old	32
40-54 years old	27
55+ years old	25

Preference share evolution by age group I

The data was divided in the four considered age groups and afterwards the preference share for each subgroup was calculated for the three surveys. As such, we can see the change in support for option A and B over the three surveys.

Age group	Option	Survey 1 (%)	Survey 2 (%)	Survey 3 (%)
17-24	Option A	59.8	64.2	56.1
25-39	Option A	68.2	67.8	65.1
40-54	Option A	61.9	67.6	59.9
55+	Option A	50.8	52.4	58.8

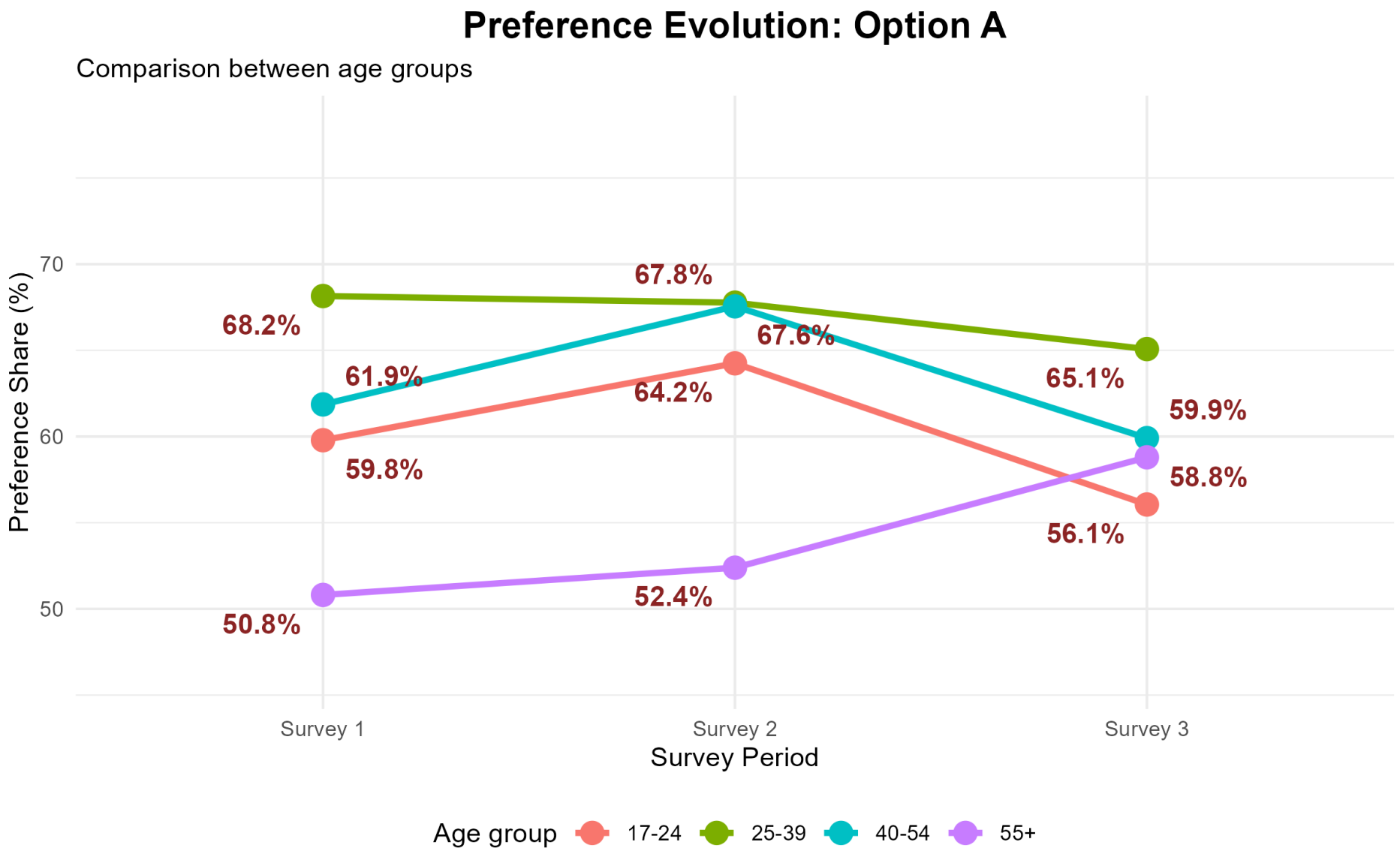
Table: Preference share evolution for option A by age group.

Preference share evolution by age group II

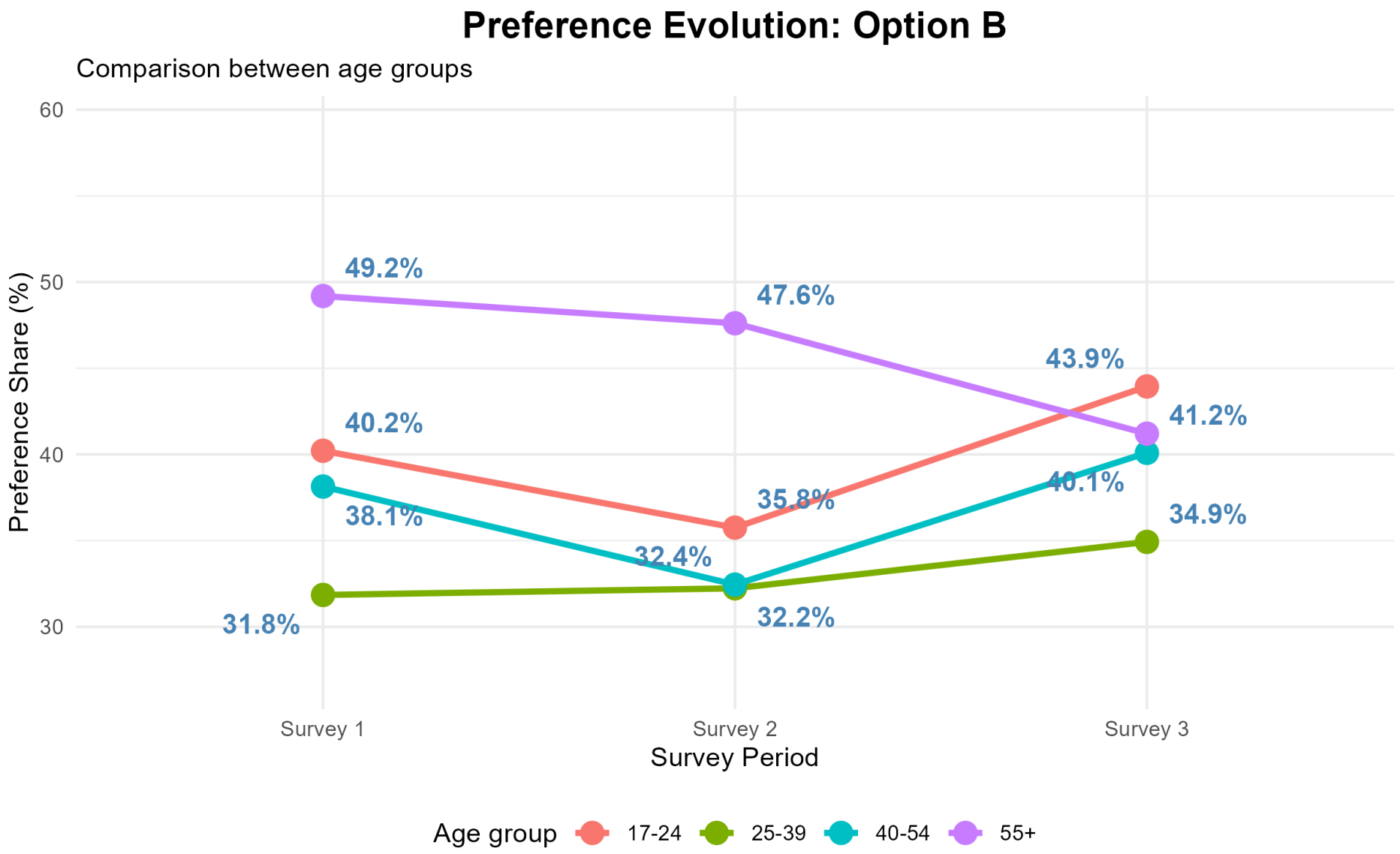
Age group	Option	Survey 1 (%)	Survey 2 (%)	Survey 3 (%)
17-24	Option B	40.2	35.8	43.9
25-39	Option B	31.8	32.2	34.9
40-54	Option B	38.1	32.4	40.1
55+	Option B	49.2	47.6	41.2

Table: Preference share evolution for option B by age group.

Evolution of preference share for A by age group



Evolution of preference share for B by age group



Bibliography I

[1] Andreas Flache, Michael Mäs, Thomas Feliciani, Edmund Chattoe-Brown, Guillaume Deffuant, Sylvie Huet, and Jan Lorenz. Models of Social Influence: Towards the Next Frontiers. *Journal of Artificial Societies and Social Simulation*, 20(4):2, 2017.

[2] GEA-ISA. Estudios efectuados durante el proceso electoral federal 2023-2024. Public Repository of the National Electoral Institute (INE), 2024. Legally mandated filing of electoral surveys.

[3] Steven F. Railsback and Volker Grimm. *Agent-Based and Individual-Based Modeling: A Practical Introduction*. Princeton University Press, Princeton ; Oxford, second edition edition, 2019.

[4] Paul E. Smaldino. *Modeling Social Behavior: Mathematical and Agent-Based Models of Social Dynamics and Cultural Evolution*. Princeton University Press, Princeton Oxford, 2023.